Nova Travis

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Education

Northwestern University **Evanston, IL**

Bachelor of Science in Mathematics and Computer Science *Expected Jun 2028*

Data Structures, Multivariable Calculus, Linear Algebra, Functional/Object-Oriented Programming, Discrete Math, Abstract Algebra, Fourier Analysis, Stochastic Differential Calculus

Research

Loop Structure within the Collatz Sequence Apr 2025 – Mar 2026

- Represented odd-step transitions with packet maps $P_n(x) = (2^n x - 1)/3$ and their compositions, deriving **parity and divisibility constraints** on arbitrary loops.
- Implemented an **inverse-prime transition framework**: for each prime $p > 3$, computed the residue $-3^{-1} \pmod{p}$ and mapped prime divisors of a to those of $3a + 1$ to **enumerate admissible chains** within hypothetical Collatz cycles.
- Developed an algorithm imposing **arithmetic and parity constraints** on solutions to the nested packet-map equation $P_N(x) = x$, excluding classes of candidate cycles and deriving an **explicit lower bound** on the odd terms required in any nontrivial cycle.

Spectrum: Dynamic Game-Theoretic Systems Mar 2026 – Present

- Formalized a **time-discounted utility model**, dynamically interpolating between the product of opposing-team population shares and a baseline utility as published information becomes stale.
- Developing switching-matrix and graph-based models of migration, equilibrium, and emergent coordination; integrating **Markov chains**, stochastic processes, **optimization**, and agent-based simulation for a campus-scale experiment.

Growth Properties of Tetration Functions Dec 2024 – Feb 2025

- Analyzed **finite-height tetration** and recursively nested exponential functions, **deriving chain-rule recurrences** for n-th derivatives and comparing growth across tower height.
- Used symbolic and numerical computation to study **critical points, asymptotic behavior, and growth-rate separation** between exponential and tetrational families.

Technical Experience & Leadership

Data Science / Machine Learning Intern, Kessler-Ellis Products Jul 2025 – Aug 2025

- Built an **XGBoost regression pipeline** in Python to predict smart-radiator valve positions from setpoint error, boiler status, and time-series indoor/outdoor temperature data, including lagged, rolling, and OHLC-derived features.
- Achieved an **RMSE of approximately 5** on held-out data; applied **exponential moving-average smoothing** and rule-based post-processing to stabilize actuator commands, and used SHAP analysis to interpret feature effects.
- Cleaned, joined, and transformed **80,000+** telemetry rows with Pandas; packaged reproducible preprocessing and training scripts for integration into an **IoT HVAC control system** associated with **30%+ energy reduction**.

Founder & President / Research Director, SVPRNOVA Feb 2026 – Present

- Founded a 60+ member **interdisciplinary research organization** building student-led research programs across several disciplines with expansion into university chapters.
- Designed SVPRNOVA's **research framework and technical platform**, organizing open-ended questions into collaborative projects, structured research pathways, researcher profiles, and tools for interdisciplinary discovery and discussion.
- Direct research strategy, project scoping, technical development, faculty partnerships, and publication planning; launched projects combining **mathematical modeling, software development, and empirical experimentation**.

Northwestern FinTech Intern Project Sep 2025 – Dec 2025

- Built an end-to-end **BTC market-data research pipeline** using Gemini trade data, with Python/C++ data ingest, C++ feature computation, pybind11 bindings, and Python inference.
- Implemented order-flow features and an **online Lasso model** to predict short-term midprice changes; evaluated signal quality through prediction-target correlation analysis.

Skills + Interests

Languages Python, Java, C++, JavaScript
Tools Pandas, NumPy, scikit-learn, XGBoost, SHAP, Git/GitHub, LaTeX
Interests Chess (2000 ELO), Taekwondo (2nd Degree Black Belt), Calisthenics, Poetry, Pencil Sketching